

Author of the PathSim simulation ecosystem and of a vertically integrated simulation stack for electronics: electromagnetic field solvers, symbolic circuit analysis, and system-level simulation. One architecture throughout: SSA compute graphs at the core of the engines, Rust cores, Python APIs, browser interfaces.

EXPERIENCE

- 2025 - present **Independent engineering consultant** · self-employed
- Custom solvers and engineering tools for client teams; commercial licensing of FastSim, SANE and RapidMoM
 - Client work including in-browser C compilation and GPU dispatch of batched dynamical systems (under NDA)
 - Open-source collaboration with the MIT Plasma Science and Fusion Center and the UK Atomic Energy Authority on tritium fuel-cycle modelling
- 2023 - 2025 **Research associate** · Institut für CMOS Design, TU Braunschweig
- Numerical modelling of electrochemical sensors for analog circuit design
 - System design and hardware for bio-impedance spectroscopy
 - EDA pipeline for RFIC passives, validated in silicon for cryogenic quantum applications
- 2021 - 2023 **Research assistant** · Institut für CMOS Design, TU Braunschweig
- High-frequency measurement calibration methods and verification hardware; taught the circuit theory seminar
- 2019 - 2020 **Intern, electrics/electronics development** · Volkswagen AG, Wolfsburg
- Low-voltage and redundant power supply systems: concept, schematic, hardware, ASIL considerations
- 2018 - 2021 **Teaching assistant** · TU Braunschweig
- Weekly independent tutorials on linear dynamical systems and circuit theory for 50+ students

EDUCATION

- 2023 - present **PhD candidate, electrical engineering (external)** · TU Ilmenau
- 2021 - 2023 **M.Sc. electrical engineering, with honors** · TU Braunschweig
Thesis (graded excellent): area optimized neural network based modelling of passive magnetic structures for chip-level integration
- 2015 - 2021 **B.Sc. electrical engineering** · TU Braunschweig
Thesis (graded excellent): order reduction of interconnect network models for efficient calculation with SPICE simulators

OPEN SOURCE & PRODUCTS

- PathSim** Python framework for hybrid system simulation. JOSS-published, 400+ GitHub stars, adopted by JSBSim, used at MIT PSFC and UKAEA for fusion fuel-cycle modelling.
- FastSim** Drop-in Rust engine for PathSim: trace-based JIT, symbolic autodiff, FMI 3.0, C99 code generation. 191x median per-step speedup over the Python engine.
- SANE** Symbolic and numeric circuit analysis in Rust with a native Verilog-A frontend; validated against ngspice and Xyce, scales past 100k parameters.

RSLAB	Sparse direct solver in pure Rust (LDLT, LU, KLU paths); benchmarked against MKL PARDISO and faer, bit-identical across thread counts.
RapidFEM / RapidMoM / RapidMesh / RapidPassives	Electromagnetic solver stack: Maxwell FEM, 2.5D MoM for planar RF passives, tetrahedral and planar meshing, browser-based RFIC layout generation.
PathView	Browser-based visual editor for PathSim models, running client-side through Pyodide. Started at MIT PSFC, now part of the PathSim organization.

PUBLICATIONS & TALKS

M. Rother, "PathSim - A System Simulation Framework", Journal of Open Source Software 10(109), 8158, 2025.
R. Delaporte-Mathurin, R. MacDonald, J. Dark, M. Rother, T. Zulfiqar, K. B. Woller, "Physics-informed tritium fuel cycle modelling workflow for fusion reactors", arXiv:2603.25751, 2026.
Talk: PathSim, IAEA Workshop on Digital Engineering for Fusion Energy Research, December 2025.

SKILLS

Programming: Python, Rust, TypeScript, C, LaTeX

Numerics: FEM and MoM electromagnetics, circuit simulation (MNA, harmonic balance, Verilog-A), ODE/DAE solvers, sparse direct solvers, model order reduction, automatic differentiation, JIT compilation

Web & tooling: SvelteKit, WebGL, WebAssembly / Pyodide, PyO3, CI pipelines

BEYOND ENGINEERING

2014 - 2021 Competition climbing coach, DAV Braunschweig: led the team from entry level to international level.